



They have 12 sensors, each sending six numbers constantly, so the AI receives 72 data points every moment. These numbers form unique patterns for each sign, like “hello” or “how are you?” The AI reads these patterns to identify the sign. Since sign language follows a different word order—like ‘cat mouse ran’ instead of ‘a cat ran after the mouse’—the AI uses natural language processing to translate signs into proper English sentences. It also learns to recognise small differences in how people sign, just as handwriting varies, making communication easier for all.

Regarding the electronics involved, Aishwarya explains, “We use two types of PCBs: a main one on the wrist and 10 custom ones on the fingertips. The main PCB includes a motion sensor, ESP32 chip, microphone, speaker, haptic motor, screen, and supporting components. Each fingertip PCB has a motion sensor, with one fingertip featuring a button. You press the button, perform sign gestures, then press it again. The gestures between presses are sent to the cloud for processing, helping the system distinguish sign language from other hand movements such as scratching.”

The gloves currently support Indian sign language, with plans to add American and British sign



Aishwarya Karnataki, Founder and CEO, Glovatrix

languages. The spoken output can be in Hindi, Marathi, English, or other languages, based on user preference. An application on the phone allows users to easily change the settings.

Discussing design challenges, Aishwarya says, “The biggest challenge in developing the app was retrieving data from the hardware—especially from two gloves simultaneously. Initially, we built separate apps for Android and iOS, but our partner, Coreco Technologies, recommended using Flutter for cross-platform development. They even created a basic Bluetooth-connected prototype for free, then later built the full app. Now, the right glove sends its data to the left glove, which

aggregates both data sets and transmits them to the phone.”

The PCBs are made in China and arrive with some surface-mount components (SMDs) pre-attached. Remaining parts, sourced from Japan, the US, Taiwan, China, and other countries, are hand-soldered in India. The outer casing is locally 3D printed, and gloves are stitched by a local tailor. All components are assembled and tested in-house.

Speaking about revenue, Aishwarya adds, “We have not sold any device yet. However, we did get support through a CSR initiative from a hotel chain called Royal Orchid Hotels, which has recently started hiring people with hearing and speech disabilities. Through their CSR, they have placed a pre-order for 10 devices as part of a paid pilot project. We will start deploying those next month.”

Looking ahead, the main goal is to enter the market via B2B sales, starting by deploying gloves to organisations. After gaining traction in select locations, the startup will seek investment to grow and expand marketing, sales, and research teams, and build a large sign language gesture database. By the end of 2026, they plan to launch directly to consumers and expand globally by adding American sign language support.

—Nidhi Agarwal

3 PARKMATE REDEFINING INDIA'S PARKING EXPERIENCE WITH INNOVATION, EFFICIENCY, AND CARE

ParkMate, a UP startup, addresses parking inefficiencies such as congestion, revenue loss, and safety through smart solutions.

Founded by Dhananjaya Bharadwaj and Abhimanyu Singh, ParkMate is a smart parking technology company addressing India's long-standing parking challenges with a

comprehensive end-to-end solution for enterprises and general users.

Officially registered in July 2021, the startup began operations in April 2022, initially targeting B2B clients, including smart cities, malls,

hospitals, and other high-traffic locations. It deployed its proprietary Trident System—a parking management platform powered by AI-based cameras, RFID sensors, and FASTag integration.